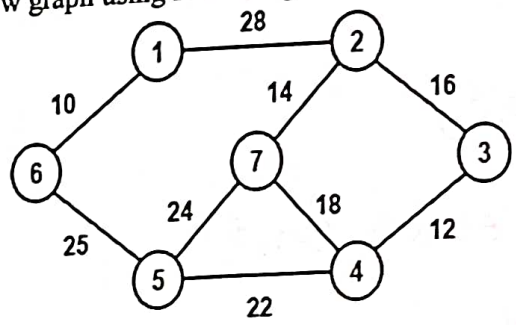
ASSIGNMENT-7

14/9/2022

6/6

4. Explain the different steps used to find the minimum cost spanning tree for the below graph using Prim's algorithm.(6 Marks)



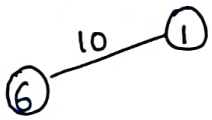
Initially,

{0, INF, INF, INF, INF, INF, INF}

= keyset {1} = mstset

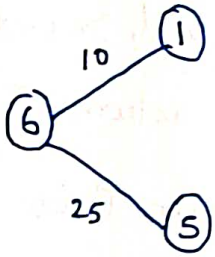
From 1, (1,2) = 28 < INF
(1,6) = 10 < INF
minimum (1,6) add mstset

keyset = {0, 28, INF, INF, INF, 10, INF} mstset = {1,6}



From 1, From 6

(1,2) = 28 (6,5) = 25 < INF



minimum (6,5) = add mstset

key set = {0, 28, INF, INF, 25, 10, INF}

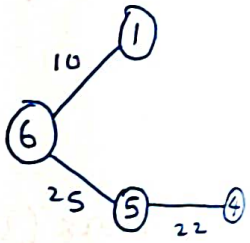
mstset = {1, 6, 5}

From 1 From 6 From 5

(1,2) = 28 now new edges (5,7) = 24 < INF

So no need further (5,4) = 22 < INF

update keyset



minimum = (5,4) add mstset

keyset = {0, 28, INF, 22, 25, 10, 24}

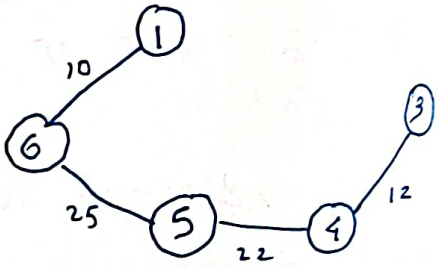
mstset = {1, 6, 5, 4}

From 1 From 5 From 4,

(1,2) = 28 (5,7) = 24 (4,7) = 18 < 24

(4,3) = 12 < INF

update keyset



minimum (4,3) add mstset

keyset = {0, 28, 12, 22, 10, 18}

mstset = {1, 6, 5, 4, 3}

From 1 | From 5 | From 4 | From 3

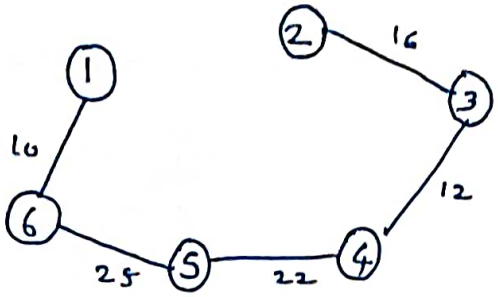
(1,2) = 28 | (5,7) = 24 | (4,7) = 18 | (3,2) = 16 < 28

update keyset

pick (3,2) minimum on mstset

keyset = {0, 16, 12, 22, 25, 10, 18}

mstset = {1, 6, 5, 4, 3, 2}



From 1 | From 3 | From 5 | From 4 | From 2

we got minimum edge to 2 so no need to consider further (1,2) = 28 | no new edge | (5,7) = 24 | (4,7) = 18 | (2,7) = 14 < 18

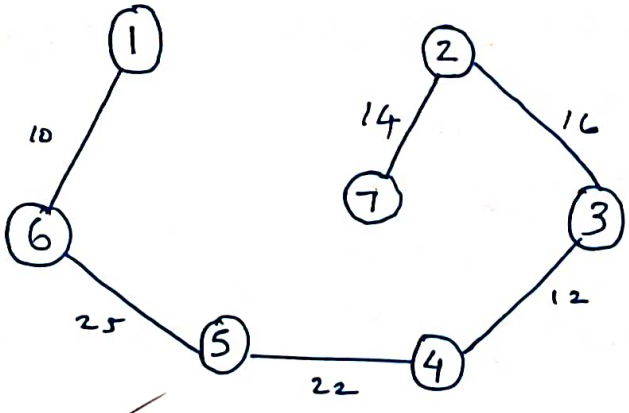
update keyset

pick (2,7) minimum on mstset

keyset = {0, 16, 12, 22, 25, 10, 14}

mstset = {1, 6, 5, 4, 3, 2, 7}

all vertices are available on mstset so stop the process



Total weight = 99